1. Implement the pure or multiplicative LCG $x_{i+1} = ax_i \bmod m$ to generate a sequence x_i, and hence the uniform random numbers u_i, with the following sets of values of a and m (take a common seed):

   (a) $a = 16807$ and $m = 2^{31} - 1$ (known as the *minimal standard generator*).

   (b) $a = 48271$ and $m = 2^{31} - 1$.

   (c) $a = 40692$ and $m = 2147483399$.

   (d) $a = 40014$ and $m = 2147483563$.

   Group the values into equidistant ranges for the values of u_i. Tabulate the proportions and draw a bar diagram for the above. What do you observe ?

   Plot the values (u_i, u_{i+1}) on a unit square. Now, zoom into the range $u_i \in [0, 0.001]$. What are your observations?

   Also, compute the autocorrelation of lags from 1 to 6. What are your observations ?

   Among the above, how do the other LCGs compare to that of the minimum standard generator, based on your observations ? Which, in your view, is the best and why?

2. Use the following linear congruence generator $x_{i+1} = ax_i \bmod 2^5$. Take the values of $a = 5$ and 7 with various initial seeds of $x_0 = 1, 2, 3$. List out the generated sequences for all the six cases along with their periods.

3. Consider the LCG to generate the sequence of random numbers u_i :

$$x_{i+1} = (2^{16} + 3)x_i \bmod 2^{31}, \qquad u_{i+1} = x_{i+1}/2^{31}.$$

   Is this a good generator, in your view? Now, evaluate the values

$$u_{i+2} - 6u_{i+1} + 9u_i.$$

   Do you notice any interesting pattern ? What is the implication for the distribution of the triplets (u_i, u_{i+1}, u_{i+2}) in the unit cube ? How many planes do you see, on which the points fall?

4. Consider the recursion:

$$U_{i+1} = (U_{i-17} - U_{i-5}).$$

   In the event that $U_i < 0$, set $U_i = U_i + 1$. Choosing seed appropriately, perform following parts.

   (a) Use linear congruence generator to generate the first 17 values of U_i.

   (b) Then generate the values of $U_{18}, U_{19}, \ldots, U_N$ for $N = 1000, 10000$, and 100000 based on the recursion above.

   (c) For each N, plot histogram. What are your observations?

   (d) For each N, plot (U_i, U_{i+1}). What are your observations?

Submission Deadline: August 08, 2026, 11:59 PM